

Grade 3 Mathematics

Մաթեմատիկա — 3-րդ դասարան

COURSE CODE	MTH-G3
PROGRAM	Elementary Mathematics · Տարրական մաթեմատիկա
LEVEL	Grade 3 · typical age 8-9
DURATION	72-84 instructional hours (approx. 66 lessons)
PREREQUISITE	Grade 2 Mathematics or equivalent: fluency within 100, place value to 1,000, arrays and equal groups.
LEADS TO	Grade 4 Mathematics (MTH-G4)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 3 is the pivot year of elementary mathematics. Two ideas of permanent consequence arrive at once: multiplication and division as operations in their own right, and fractions as numbers rather than as pieces of a picture. Students also meet area as a measurable quantity, extend arithmetic to larger numbers, and begin to solve genuinely two-step problems. A student who leaves Grade 3 with secure multiplication facts and a number-line understanding of fractions is set up for everything that follows.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Two commitments define how Վարժարան teaches Grade 3. First, multiplication facts are built through structure — skip counting, arrays, the distributive property, and derived facts — and only then automatized; drilling before understanding produces brittle recall that fails under multi-digit load. Second, fractions are taught on the number line from the very first lesson, not only as shaded regions. The pizza model alone is the single largest cause of fraction failure in Grades 5-7, because it never shows the student that $\frac{3}{4}$ is a number with a location.

PLACEMENT AND READINESS INDICATORS

Ընդունելություն և պատրաստվածություն

The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Adds and subtracts within 100 fluently	Multi-digit work and two-step problems both depend on this; remediate before Unit 4 if weak.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Builds and interprets a rectangular array	Direct prerequisite for multiplication; if absent, begin with Grade 2 Unit 5.
Skip counts by 2s, 5s, and 10s from any start	Provides the first multiplication facts and the rhythm for the rest.
Partitions a shape into equal shares and names one share	The conceptual entry point for fractions.

COURSE LEARNING OUTCOMES

Ուսումնական արդյունքներ

On completion of this course the student will be able to:

- 1 Interpret products and quotients of whole numbers in terms of equal groups, arrays, and area.
- 2 Know all products of two one-digit numbers from memory by the end of the course.
- 3 Apply the commutative, associative, and distributive properties as strategies for multiplication.
- 4 Understand division as an unknown-factor problem and solve division within 100.
- 5 Solve two-step word problems using all four operations and assess the reasonableness of answers.
- 6 Understand a fraction $1/b$ as one part of a whole partitioned into b equal parts, and a/b as a copies of $1/b$.
- 7 Represent fractions on a number line and generate and explain equivalent fractions.
- 8 Compare fractions with the same numerator or the same denominator and justify the comparison.
- 9 Understand area as covering with unit squares, and relate area to multiplication and addition.
- 10 Solve problems involving perimeter, elapsed time, mass, liquid volume, and scaled graphs.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 8-9 LESSONS

Meaning of Multiplication

Բազմապատկման իմաստը

TOPICS COVERED

- Interpreting 5×7 as five groups of seven
- Arrays, area models, and equal-groups pictures
- The multiplication symbol and reading equations aloud
- The commutative property and why it halves the work
- Facts for 2, 5, and 10 from skip counting
- Facts for 0 and 1 and why they behave as they do
- The distributive property concretely: 7×6 as 5×6 plus 2×6
- Deriving 3s, 4s, 6s, and 8s from known facts
- Multiplying by multiples of 10

LEARNING OBJECTIVES

- 1 Write a multiplication equation for an equal-groups or array situation and interpret each factor.
- 2 Use a known fact to derive an unknown one via the distributive property.
- 3 Explain why $4 \times 7 = 7 \times 4$ using an array, in words.
- 4 Multiply a one-digit number by a multiple of 10 using place-value reasoning.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Interprets both factors correctly in context in 9 of 10 trials.
- Derives at least three unknown facts from known facts, explaining the decomposition.
- Reaches 80% accuracy on facts for 0, 1, 2, 5, and 10 with under 4 seconds per item.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
'Multiplication makes bigger' is adopted as a rule.	True for whole numbers above 1, false everywhere else. Never state it. Preempt it by including $\times 1$ and $\times 0$ early and by noting aloud that this will change with fractions.
The child does not know which factor is the group count and which is the group size.	Insist on a consistent convention in stories and then show that the product is the same either way — the <i>interpretation</i> differs even though the answer does not.

TEACHING NOTE

Introduce the area model here, not later. It is the same model that carries multi-digit multiplication in Grade 4, fraction multiplication in Grade 5, and binomial multiplication in Algebra I.

UNIT 2 • 8-9 LESSONS**Division and the Multiplication Facts**

Բաժանում և բազմապատկման աղյուսակ

TOPICS COVERED

- Partitive division: sharing equally among a known number of groups
- Quotitive division: measuring out groups of a known size
- The division symbol and notation
- Division as an unknown-factor problem
- Fact families linking multiplication and division
- Dividing by 1 and the impossibility of dividing by 0
- Systematic fluency work on all facts to 10×10
- Solving one-step multiplication and division word problems

LEARNING OBJECTIVES

- 1 Distinguish sharing situations from measuring situations and model each correctly.
- 2 Rewrite a division problem as a missing-factor multiplication problem.
- 3 Know from memory all products of two one-digit numbers.
- 4 Explain why division by zero has no answer, using an equal-groups argument.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers 60 mixed facts to 10×10 with 90% accuracy in 3 minutes.
- Writes the complete fact family for a given triple in 5 of 5 trials.
- Correctly models both division meanings for the same numerical problem.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Division is assumed to be commutative: $12 \div 3$ treated as $3 \div 12$.	Contrast with subtraction, which the student already knows is not commutative. Model both with objects and let the child see one is impossible.
Fact fluency is pursued through timed drill alone and stalls around the 6s, 7s, and 8s.	Switch to derived-fact strategies: 7×8 as 7×4 doubled, or 8×8 minus 8. Structure unlocks what drill cannot.

TEACHING NOTE

Automatic recall of the multiplication facts by the end of Grade 3 is the strongest single predictor of Grade 5-7 success. Fluency work continues in every lesson for the rest of the year, even after this unit closes.

UNIT 3 • 10-11 LESSONS

Fractions as Numbers

Կոտորակները որպես թվեր

TOPICS COVERED

- Partitioning a whole into b equal parts; the unit fraction $1/b$
- Building a/b as a copies of the unit fraction
- Fractions on the number line between 0 and 1
- Fractions greater than 1 on the number line
- Whole numbers as fractions ($3 = 3/1$, $4/4 = 1$)
- Equivalent fractions through partitioning and on the number line
- Recognizing simple equivalences: $1/2 = 2/4 = 3/6$
- Comparing fractions with the same denominator
- Comparing fractions with the same numerator
- Comparing to the benchmark $1/2$
- Why comparisons require the same whole

LEARNING OBJECTIVES

- 1 Locate any fraction with denominator 2, 3, 4, 6, or 8 on a number line and explain the placement.
- 2 Explain a/b as a copies of $1/b$ using a drawing and a number line.
- 3 Generate an equivalent fraction and justify it with a partitioned model.
- 4 Compare two fractions with the same numerator and explain why the larger denominator gives the smaller fraction.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Places fractions correctly on a number line in 9 of 10 trials.
- Justifies an equivalence with a model, not a rule, in 4 of 5 trials.
- Compares same-numerator fractions correctly in 9 of 10 trials with reasoning.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A larger denominator is assumed to mean a larger fraction, because $8 > 4$.	The denominator names the size of the piece, not the count. Fold identical paper strips into 4 and into 8 and compare a single piece from each — physically.
Fractions are seen only as shaded parts of a shape, never as numbers.	Every fraction lesson in this unit must include a number-line representation. This is non-negotiable at Վարժարան.
Comparisons are made across different wholes ($1/2$ of a large pizza vs. $3/4$ of a small one).	State explicitly and often: comparisons are only valid when the whole is the same. Use a deliberate counterexample.

TEACHING NOTE

This is the most important unit in the elementary curriculum. Grade 5–7 fraction failure is almost always traceable to a Grade 3 unit taught in two weeks with pizzas. Give it the time it needs.

UNIT 4 • 6–7 LESSONS**Area and Perimeter**

Մակերես և պարագիծ

TOPICS COVERED

- Area as covering a region with unit squares
- Measuring area by counting unit squares
- Relating area to multiplication of side lengths
- Area of rectilinear figures by decomposition
- The distributive property seen as area: $a \times (b + c)$
- Perimeter as distance around
- Distinguishing area from perimeter
- Rectangles with equal area but different perimeter, and vice versa

LEARNING OBJECTIVES

- 1 Find the area of a rectangle by tiling and by multiplying, and show these agree.
- 2 Decompose a rectilinear figure into rectangles and add the areas.
- 3 Find an unknown side length given the perimeter and the other sides.
- 4 Construct two rectangles with the same area and different perimeters.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes rectangle area correctly with units in 9 of 10 trials.
- Decomposes an L-shaped figure and finds total area in 4 of 5 trials.
- Explains the difference between area and perimeter in the student's own words.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Area and perimeter formulas are confused with each other.	Do not lead with formulas. Have the student walk the perimeter with a finger and cover the area with squares. The physical distinction precedes the symbolic one.
Square units are reported as linear units.	Require correct units on every answer from the first lesson; treat a missing 'square' as an incorrect answer.

TEACHING NOTE

Area is where multiplication becomes geometric. Explicitly show that the area model of Unit 1 and the area of Unit 4 are the same picture.

UNIT 5 • 7-8 LESSONS

Multi-Digit Arithmetic and Two-Step Problems

Բազմանիշ թվարանություն և երկքայլ խնդիրներ

TOPICS COVERED

- Fluent addition and subtraction within 1,000
- Rounding to the nearest 10 and 100
- Estimating to check reasonableness
- Multiplying one-digit numbers by multiples of 10
- Two-step word problems with all four operations
- Representing unknowns with a letter or a box
- Identifying arithmetic patterns in the addition and multiplication tables
- Explaining a pattern using properties of operations

LEARNING OBJECTIVES

- 1 Solve two-step word problems and write an equation with a symbol for the unknown.
- 2 Round to the nearest 10 or 100 and use rounding to assess reasonableness.
- 3 Identify and explain a pattern in a multiplication table (e.g., why all products of 4 are even).

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves two-step problems correctly in 8 of 10 trials with an equation shown.
- Rounds correctly in 9 of 10 trials, including at the midpoint.
- Explains one table pattern using a property of operations.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
In a two-step problem, only the first step is completed.	Require the student to restate the question in their own words before computing, and to check the answer against that restatement afterward.
Rounding is done by looking at the wrong digit.	Use a number line and ask 'which ten is it closer to?' rather than teaching a digit rule.

TEACHING NOTE

Two-step problems are where reading comprehension and mathematics meet. Teachers should read the problem aloud with the student and model the underlining of the question.

UNIT 6 • 6-7 LESSONS

Measurement, Data, and Geometry

Հափոււմ, տվյալներ և երկրաչափություն

TOPICS COVERED

- Telling time to the minute; elapsed-time problems on a number line
- Measuring and estimating liquid volume and mass in metric units
- Solving one-step problems involving mass and volume
- Scaled picture graphs and scaled bar graphs
- Line plots with halves and fourths of an inch
- Classifying quadrilaterals by their attributes
- Rhombuses, rectangles, and squares as related categories
- Partitioning shapes into equal areas and naming the share

LEARNING OBJECTIVES

- 1 Solve an elapsed-time problem using a number line.
- 2 Read a scaled bar graph where one square represents more than one unit.
- 3 Classify quadrilaterals by attributes and explain why a square is also a rectangle.
- 4 Measure lengths to the nearest quarter inch and display them on a line plot.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves elapsed-time problems correctly in 8 of 10 trials.
- Interprets a scaled graph correctly in 4 of 5 trials.
- Correctly explains one shape-category inclusion relationship.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
On a scaled graph each square is read as one unit.	Require the student to state the scale aloud before reading any value.
A square is not recognized as a rectangle.	Category inclusion is genuinely hard at this age. Use attribute checklists rather than appearance.

TEACHING NOTE

Line plots with fractional measurements deliberately reunite Units 3 and 6. Point this out to the student — connections should be named, not left to chance.

ASSESSMENT AND MASTERY POLICY

Գնահատում և յուրացման քաղաքականություն

ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Grade 2 exit skills and readiness for multiplication; produces the initial academic profile.
Unit Check	End of each unit · 20 min	Unit mastery criteria; includes at least one 'explain your reasoning' item.
Multiplication Fluency Probe	Weekly from Unit 2 onward · 3 min	All facts to 10×10 ; tracked as a running chart the student can see.
Fraction Diagnostic	During and after Unit 3	Number-line placement, equivalence, and comparison; the highest-stakes internal check of the year.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3; weighted toward multiplication meaning and fractions.
End-of-Course Assessment	After Unit 6	All course outcomes; determines Grade 4 placement and whether fraction remediation is needed.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

Ուսումնական տեմպ

TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 50 min	Full course in approximately 45–50 weeks.
Accelerated	2 lessons/week	Full course in approximately 24–28 weeks.
Intensive / Catch-Up	2–3 lessons/week	18–22 weeks; Units 1–3 prioritized.
Fractions Intensive	2 lessons/week, 6 weeks	Unit 3 only — offered as a standalone remediation for students in Grades 4–6 with fraction gaps.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMework AND INDEPENDENT PRACTICE

Տնային աշխատանք և ինքնուրույն պարապմունք

Twenty minutes per lesson. Every assignment includes a short multiplication-fact component (2–3 minutes) that continues from Unit 2 through the end of the course, regardless of the current topic — this is what produces automaticity by June. The remainder is three to five problems on current content plus one review problem. Word problems appear in every assignment.

PROGRESS MILESTONES REPORTED TO PARENTS

Ուսումնական նվաճումներ

These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Interprets multiplication and division in terms of equal groups and arrays.
- ◆ Knows all multiplication facts to 10×10 from memory.
- ◆ Understands division as an unknown-factor problem.
- ◆ Places fractions on a number line and explains a/b as a copies of $1/b$.
- ◆ Generates and justifies equivalent fractions.
- ◆ Compares fractions with the same numerator or denominator, with reasoning.
- ◆ Relates area to multiplication and distinguishes it from perimeter.
- ◆ Solves two-step word problems with all four operations.
- ◆ Completed Grade 3 Mathematics — ready for Grade 4.

MATERIALS AND RESOURCES

Նյութեր և ռեսուրսներ

- Fraction strips and fraction number-line kits (physical set at home, digital on the shared board).
- Square tiles for area work; grid paper.
- Multiplication fact chart — used as a pattern-discovery tool, not as a crutch during fluency work.
- Վարժարան Grade 3 problem bank — approximately 1,000 items, with a dedicated fraction sub-bank of 250.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.